

DATA620_Project3_Gender-Classifer

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0.1 DATA 620 Project Three Instructions

- Split the Names Corpus into three subsets: 500 words for the test set, 500 words for the dev-test set, and the remaining 6900 words for the training set.
- Starting with the example name gender classifier, make incremental improvements.
- Use the dev-test set to check your progress.
- Check its final performance on the test set.
- How does the performance on the test set compare to the performance on the dev-test set?
- Is this what you'd expect?

0.1.1 Executive Summary

Following the Bird et al (2009) chapter 6 guidance, a gender classification model was built using the NLTK names corpus and a Naive Bayes classifier, following a Supervised Learning workflow. The dataset of labeled names was split into training, development, and test sets, and feature engineering was applied to convert names into feature extractors based on known patterns. The model was trained on the training set and iteratively improved using the dev set. Using just the first letter of names, I started with a 64.6% performance rate on the training data and a 65.8% rate on the dev_set data. The model incrementally improved over three feature extractor improvements (77%, 80.4% and up to 82.2%). However, on unseen test set, performance was only 67%, indicating that it does not generalize well. I speculate that this is a case of overfitting. This could be caused by being too specific in the extraction features, as seen in the Slepian & Galinsky (2016) features, or by features involving the last three letters of the names. This outcome highlights the importance of proper data splitting, avoiding overfitting during feature design, and ensuring that evaluation on truly unseen data reflects real-world performance.

0.1.2 MODULE 1: Feature Extractor.

The feature extractor builds a model (a function) that turns raw text (a word) into a labeled feature that a machine learning model or other program can use for analysis or prediction. According to the Bird et al. (2009) text, the feature extractor decides which features of the input are relevant and how to encode them. Bird et al. (2009) use the example of extracting the last letter of the word 'Shrek'. Below, I modified that example to show the feature extractor returning the value of the first letter of a name, 'Mary'. Note that the function strips case sensitivity.

```
[1]: def gender_features(word):
      word = word.lower().strip()

      return {
          'first_letter': word[0]}

gender_features('Mary')
```

```
[1]: {'first_letter': 'm'}
```

0.1.3 MODULE 2: Load the names Data from the nltk library.

According to the Bird et al. (2009) text, the next step is to prepare a list of examples and corresponding class labels. The nltk library contains lists of known male and female names in text files. The labeled_names object is created, and it iterates over each name in names.words('male.txt'), and makes a pair (name, 'male') so every name gets labeled as 'male', and does the same for the females. The names are selected randomly and set to a seed for replicability.

```
[2]: import nltk
nltk.download('names', quiet=True)
from nltk.corpus import names
labeled_names = [(name, 'male') for name in names.words('male.txt')] +
                 [(name, 'female') for name in names.words('female.txt')]

import random
random.seed(100)
random.shuffle(labeled_names)
```

0.1.4 MODULE 3: Feature Set Creation.

According to the Bird et al. (2009) text, the next step is to create a feature set, which is a dictionary that maps from feature names to their values. The featuresets organize the “names” data into a format that the classifier can look for patterns. The featuresets object iterates over each (n, gender) pair in labeled_names, converts the name n into features using gender_features(n), and stores those features with the same gender.

```
[3]: featuresets = [(gender_features(n), gender) for (n, gender) in labeled_names]

for item in featuresets[:5]:
    print(item)

({'first_letter': 'p'}, 'male')
({'first_letter': 'b'}, 'male')
({'first_letter': 'p'}, 'female')
({'first_letter': 'f'}, 'female')
({'first_letter': 'c'}, 'male')
```

0.1.5 MODULE 4: Split the Names Corpus: Training, Dev-Test, Test Subsets

In the following module, I split the data into three subsets: 500 words for the test set, 500 words for the dev-test set, and the remaining 6900 words for the training set. The training set is used to train the model, and the dev-test set is used to perform error analysis. The test set serves in our final evaluation of the system.

```
[4]: test_names = labeled_names[:500]
     dev_names = labeled_names[500:1000]
     train_names = labeled_names[1000:]
```

0.1.6 MODULE 5: Convert Data to Feature Sets

Each object below takes each name and turns it into features, then pairs those features with the correct gender so the model can learn from them.

```
[5]: train_set = [(gender_features(n), g) for (n, g) in train_names]
     dev_set   = [(gender_features(n), g) for (n, g) in dev_names]
     test_set  = [(gender_features(n), g) for (n, g) in test_names]
```

0.1.7 MODULE 6: New ‘naive Bayes’ Classifier Creation Tested on Training Data

The training set is used to train a new “naive Bayes” classifier.

```
[6]: classifier = nltk.NaiveBayesClassifier.train(train_set)

     print(f"{nltk.classify.accuracy(classifier, train_set):.3f}")
```

0.646

0.1.8 MODULE 7: New ‘naive Bayes’ Classifier Creation Tested on Dev-Set Data

The code below takes the naive Bayes classifier, which is trained on the training data, then tested on the dev_set data.

```
[7]: print(nltk.classify.accuracy(classifier, dev_set))
```

0.658

0.1.9 MODULE 8: Most Informative Features

In this module, I examine the classifier (tested on the training data) to determine which features it found most effective for distinguishing the names’ genders. The interpretation is that if a male has a name that starts with a ‘w’, it is 4.3 times more likely to be a male over a female name. If it begins with a ‘k’, it is twice as likely to be a female name.

```
[8]: classifier.show_most_informative_features(5)
```

Most Informative Features

first_letter = 'w'	male : female =	4.3 : 1.0
first_letter = 'q'	male : female =	2.9 : 1.0

first_letter = 'x'	male : female =	2.3 : 1.0
first_letter = 'k'	female : male =	2.3 : 1.0
first_letter = 'h'	male : female =	2.1 : 1.0

0.1.10 MODULE 9: Diagnose Errors

This output shows the predicted names versus the actual classifications from the dev_test corpus. For example, the model predicted Heidi to be a male name, but it is actually a female name. With understanding individual error cases where the model predicted the wrong label, I can try to determine what additional pieces of information would allow it to make the right decision, and adjust the feature set accordingly.

```
[9]: errors = []
for (name, tag) in dev_names:
    guess = classifier.classify(gender_features(name))
    if guess != tag:
        errors.append( (tag, guess, name) )

for (tag, guess, name) in sorted(errors):
    print('correct={:<8} guess={:<8s} name={:<30}'.format(tag, guess, name))
```

correct=female	guess=male	name=Heidi
correct=female	guess=male	name=Helaina
correct=female	guess=male	name=Helyn
correct=female	guess=male	name=Hermina
correct=female	guess=male	name=Hermine
correct=female	guess=male	name=Hester
correct=female	guess=male	name=Hinda
correct=female	guess=male	name=Winifred
correct=female	guess=male	name=Wynnie
correct=female	guess=male	name=Yoko
correct=male	guess=female	name=Addie
correct=male	guess=female	name=Albrecht
correct=male	guess=female	name=Aleck
correct=male	guess=female	name=Alessandro
correct=male	guess=female	name=Alston
correct=male	guess=female	name=Anatollo
correct=male	guess=female	name=Andie
correct=male	guess=female	name=Angel
correct=male	guess=female	name=Ariel
correct=male	guess=female	name=Artur
correct=male	guess=female	name=Ashby
correct=male	guess=female	name=Aubert
correct=male	guess=female	name=Barn
correct=male	guess=female	name=Barny
correct=male	guess=female	name=Barrie
correct=male	guess=female	name=Bartel
correct=male	guess=female	name=Barthel

correct=male	guess=female	name=Bartlet
correct=male	guess=female	name=Bentley
correct=male	guess=female	name=Bertie
correct=male	guess=female	name=Brett
correct=male	guess=female	name=Brook
correct=male	guess=female	name=Chaddy
correct=male	guess=female	name=Chan
correct=male	guess=female	name=Charles
correct=male	guess=female	name=Chas
correct=male	guess=female	name=Christof
correct=male	guess=female	name=Christofer
correct=male	guess=female	name=Clarance
correct=male	guess=female	name=Clayborn
correct=male	guess=female	name=Constantin
correct=male	guess=female	name=Craig
correct=male	guess=female	name=Cristopher
correct=male	guess=female	name=Cy
correct=male	guess=female	name=Darian
correct=male	guess=female	name=Davidde
correct=male	guess=female	name=Davy
correct=male	guess=female	name=Demetre
correct=male	guess=female	name=Dennie
correct=male	guess=female	name=Derick
correct=male	guess=female	name=Derrek
correct=male	guess=female	name=Derrick
correct=male	guess=female	name=Dov
correct=male	guess=female	name=Dudley
correct=male	guess=female	name=Durante
correct=male	guess=female	name=Dustin
correct=male	guess=female	name=Edouard
correct=male	guess=female	name=Elwyn
correct=male	guess=female	name=Englebart
correct=male	guess=female	name=Ephraim
correct=male	guess=female	name=Ephrem
correct=male	guess=female	name=Fairfax
correct=male	guess=female	name=Forester
correct=male	guess=female	name=Franky
correct=male	guess=female	name=Fraser
correct=male	guess=female	name=Gabriele
correct=male	guess=female	name=Garold
correct=male	guess=female	name=Georgia
correct=male	guess=female	name=Gerold
correct=male	guess=female	name=Gerrit
correct=male	guess=female	name=Giancarlo
correct=male	guess=female	name=Gibb
correct=male	guess=female	name=Gil
correct=male	guess=female	name=Gomer
correct=male	guess=female	name=Ike

correct=male	guess=female	name=Ingmar
correct=male	guess=female	name=Jarrett
correct=male	guess=female	name=Jarvis
correct=male	guess=female	name=Jeb
correct=male	guess=female	name=Jessee
correct=male	guess=female	name=Jimmy
correct=male	guess=female	name=Karl
correct=male	guess=female	name=Kermie
correct=male	guess=female	name=Kin
correct=male	guess=female	name=King
correct=male	guess=female	name=Klaus
correct=male	guess=female	name=Kris
correct=male	guess=female	name=Langston
correct=male	guess=female	name=Leigh
correct=male	guess=female	name=Lev
correct=male	guess=female	name=Linoel
correct=male	guess=female	name=Lonnie
correct=male	guess=female	name=Loren
correct=male	guess=female	name=Ludvig
correct=male	guess=female	name=Maddie
correct=male	guess=female	name=Madison
correct=male	guess=female	name=Manfred
correct=male	guess=female	name=Marc
correct=male	guess=female	name=Markus
correct=male	guess=female	name=Martin
correct=male	guess=female	name=Marve
correct=male	guess=female	name=Marvin
correct=male	guess=female	name=Marwin
correct=male	guess=female	name=Mauritz
correct=male	guess=female	name=Maynard
correct=male	guess=female	name=Michal
correct=male	guess=female	name=Moises
correct=male	guess=female	name=Morgan
correct=male	guess=female	name=Morty
correct=male	guess=female	name=Myles
correct=male	guess=female	name=Nathanael
correct=male	guess=female	name=Ned
correct=male	guess=female	name=Neron
correct=male	guess=female	name=Nickolas
correct=male	guess=female	name=Nikki
correct=male	guess=female	name=Noah
correct=male	guess=female	name=Octavius
correct=male	guess=female	name=Oliver
correct=male	guess=female	name=Osbourn
correct=male	guess=female	name=Oscar
correct=male	guess=female	name=Pablo
correct=male	guess=female	name=Palmer
correct=male	guess=female	name=Pascale

correct=male	guess=female	name=Patsy
correct=male	guess=female	name=Percival
correct=male	guess=female	name=Percy
correct=male	guess=female	name=Prasad
correct=male	guess=female	name=Raimund
correct=male	guess=female	name=Reagan
correct=male	guess=female	name=Rickard
correct=male	guess=female	name=Rik
correct=male	guess=female	name=Ripley
correct=male	guess=female	name=Ronny
correct=male	guess=female	name=Ruby
correct=male	guess=female	name=Sancho
correct=male	guess=female	name=Sarge
correct=male	guess=female	name=Saul
correct=male	guess=female	name=Sergei
correct=male	guess=female	name=Shaw
correct=male	guess=female	name=Sheldon
correct=male	guess=female	name=Shep
correct=male	guess=female	name=Shepherd
correct=male	guess=female	name=Shepperd
correct=male	guess=female	name=Slade
correct=male	guess=female	name=Smith
correct=male	guess=female	name=Sparky
correct=male	guess=female	name=Spence
correct=male	guess=female	name=Steffen
correct=male	guess=female	name=Sterne
correct=male	guess=female	name=Steve
correct=male	guess=female	name=Tabb
correct=male	guess=female	name=Taber
correct=male	guess=female	name=Tailor
correct=male	guess=female	name=Tally
correct=male	guess=female	name=Tarzan
correct=male	guess=female	name=Tedmund
correct=male	guess=female	name=Terence
correct=male	guess=female	name=Thaddius
correct=male	guess=female	name=Theobald
correct=male	guess=female	name=Theodore
correct=male	guess=female	name=Timothy
correct=male	guess=female	name=Toddie
correct=male	guess=female	name=Tom
correct=male	guess=female	name=Tommy
correct=male	guess=female	name=Torr
correct=male	guess=female	name=Towny
correct=male	guess=female	name=Trevar
correct=male	guess=female	name=Trey
correct=male	guess=female	name=Tudor
correct=male	guess=female	name=Val
correct=male	guess=female	name=Zacharia

0.1.11 MODULE 10: Gender classifier improvement #1.

In the Bird et al. (2009) text, they discuss classifying names ending in a, e and i to be likely associated with females, while names ending in k, o, r, s and t are likely to be associated with males. This is given in the text.

```
[10]: def gender_features(word):
      word = word.lower().strip()

      return {
          'ends_with_aeiy': word[-1] in 'aeiy',
          'ends_with_korst': word[-1] in 'korst',
          'ends_with_yn': word[-2] in 'yn'
      }

      print(gender_features("Mary"))

{'ends_with_aeiy': True, 'ends_with_korst': False, 'ends_with_yn': False}
```

```
[11]: train_set = [(gender_features(n), gender) for (n, gender) in train_names]
      dev_set = [(gender_features(n), gender) for (n, gender) in dev_names]
      classifier = nltk.NaiveBayesClassifier.train(train_set)
      print(nltk.classify.accuracy(classifier, dev_set))
```

0.77

```
[12]: classifier.show_most_informative_features(5)
```

Most Informative Features

ends_with_korst = True	male : female =	5.9 : 1.0
ends_with_aeiy = False	male : female =	3.5 : 1.0
ends_with_yn = True	female : male =	2.8 : 1.0
ends_with_aeiy = True	female : male =	2.7 : 1.0
ends_with_korst = False	female : male =	1.3 : 1.0

```
[13]: errors = []
      for (name, tag) in dev_names:
          guess = classifier.classify(gender_features(name))
          if guess != tag:
              errors.append( (tag, guess, name) )

      for (tag, guess, name) in sorted(errors):
          print('correct={:<8} guess={:<8s} name={:<30}'.format(tag, guess, name))
```

correct=female	guess=male	name=Abigail
correct=female	guess=male	name=Abigael
correct=female	guess=male	name=Adriaens
correct=female	guess=male	name=Alis
correct=female	guess=male	name=Allis
correct=female	guess=male	name=Alys
correct=female	guess=male	name=Anabel

correct=female	guess=male	name=Anais
correct=female	guess=male	name=Arabel
correct=female	guess=male	name=Arleen
correct=female	guess=male	name=Ashleigh
correct=female	guess=male	name=Avrit
correct=female	guess=male	name=Barb
correct=female	guess=male	name=Clair
correct=female	guess=male	name=Con
correct=female	guess=male	name=Dael
correct=female	guess=male	name=Delilah
correct=female	guess=male	name=Denys
correct=female	guess=male	name=Doreen
correct=female	guess=male	name=Easter
correct=female	guess=male	name=Elisabet
correct=female	guess=male	name=Eran
correct=female	guess=male	name=Evangelin
correct=female	guess=male	name=Gladys
correct=female	guess=male	name=Gwendolin
correct=female	guess=male	name=Hester
correct=female	guess=male	name=Isabeau
correct=female	guess=male	name=Joyan
correct=female	guess=male	name=Krystal
correct=female	guess=male	name=Lilias
correct=female	guess=male	name=Lynnett
correct=female	guess=male	name=Madel
correct=female	guess=male	name=Marabel
correct=female	guess=male	name=Maren
correct=female	guess=male	name=Maridel
correct=female	guess=male	name=Maris
correct=female	guess=male	name=Maryjo
correct=female	guess=male	name=Maud
correct=female	guess=male	name=Meris
correct=female	guess=male	name=Mikako
correct=female	guess=male	name=Orel
correct=female	guess=male	name=Robbin
correct=female	guess=male	name=Rochell
correct=female	guess=male	name=Ruth
correct=female	guess=male	name=Scarlet
correct=female	guess=male	name=Scarlett
correct=female	guess=male	name=Sibel
correct=female	guess=male	name=Storm
correct=female	guess=male	name=Suzan
correct=female	guess=male	name=Tamiko
correct=female	guess=male	name=Tarah
correct=female	guess=male	name=Vivyan
correct=female	guess=male	name=Winifred
correct=female	guess=male	name=Yoko
correct=male	guess=female	name=Addie

correct=male	guess=female	name=Andie
correct=male	guess=female	name=Ashby
correct=male	guess=female	name=Barny
correct=male	guess=female	name=Barrie
correct=male	guess=female	name=Bentley
correct=male	guess=female	name=Bertie
correct=male	guess=female	name=Chaddy
correct=male	guess=female	name=Clarence
correct=male	guess=female	name=Cy
correct=male	guess=female	name=Davidde
correct=male	guess=female	name=Davy
correct=male	guess=female	name=Demetre
correct=male	guess=female	name=Dennie
correct=male	guess=female	name=Dudley
correct=male	guess=female	name=Durante
correct=male	guess=female	name=Elwyn
correct=male	guess=female	name=Franky
correct=male	guess=female	name=Gabriele
correct=male	guess=female	name=Georgia
correct=male	guess=female	name=Hale
correct=male	guess=female	name=Horace
correct=male	guess=female	name=Ike
correct=male	guess=female	name=Jessee
correct=male	guess=female	name=Jimmy
correct=male	guess=female	name=Kermie
correct=male	guess=female	name=King
correct=male	guess=female	name=Lonnie
correct=male	guess=female	name=Maddie
correct=male	guess=female	name=Marve
correct=male	guess=female	name=Morty
correct=male	guess=female	name=Nikki
correct=male	guess=female	name=Pascale
correct=male	guess=female	name=Patsy
correct=male	guess=female	name=Percy
correct=male	guess=female	name=Raimund
correct=male	guess=female	name=Ripley
correct=male	guess=female	name=Ronny
correct=male	guess=female	name=Ruby
correct=male	guess=female	name=Sarge
correct=male	guess=female	name=Sergei
correct=male	guess=female	name=Slade
correct=male	guess=female	name=Sparky
correct=male	guess=female	name=Spence
correct=male	guess=female	name=Sterne
correct=male	guess=female	name=Steve
correct=male	guess=female	name=Tally
correct=male	guess=female	name=Tedmund
correct=male	guess=female	name=Terence

correct=	male	guess=female	name=Theodore
correct=	male	guess=female	name=Timothy
correct=	male	guess=female	name=Toddie
correct=	male	guess=female	name=Tommy
correct=	male	guess=female	name=Towny
correct=	male	guess=female	name=Trey
correct=	male	guess=female	name=Waverly
correct=	male	guess=female	name=Whitby
correct=	male	guess=female	name=Wiley
correct=	male	guess=female	name=Winnie
correct=	male	guess=female	name=Yancey
correct=	male	guess=female	name=Zacharia

0.1.12 MODULE 11: Gender classifier improvement #3.

The following improvements address the errors and look for possible two and three last letters patterns, the first letter and length of the names.

```
[14]: def gender_features(word):
      word = word.lower()
      return {
          'last_letter': word[-1:],
          'last_two_letters': word[-2:],
          'last_three_letters': word[-3:],
          'first_letter': word[0],
          'length': len(word),
      }
      print(gender_features("Mary"))
```

```
{'last_letter': 'y', 'last_two_letters': 'ry', 'last_three_letters': 'ary',
'first_letter': 'm', 'length': 4}
```

```
[15]: train_set = [(gender_features(n), gender) for (n, gender) in train_names]
      dev_set = [(gender_features(n), gender) for (n, gender) in dev_names]
      classifier = nltk.NaiveBayesClassifier.train(train_set)
      print(nltk.classify.accuracy(classifier, dev_set))
```

0.804

```
[16]: classifier.show_most_informative_features(5)
```

Most Informative Features

last_two_letters = 'na'	female : male	=	92.8 : 1.0
last_two_letters = 'ia'	female : male	=	88.7 : 1.0
last_two_letters = 'la'	female : male	=	71.0 : 1.0
last_letter = 'a'	female : male	=	34.9 : 1.0
last_two_letters = 'sa'	female : male	=	34.8 : 1.0

```
[17]: errors = []
      for (name, tag) in dev_names:
          guess = classifier.classify(gender_features(name))
          if guess != tag:
              errors.append( (tag, guess, name) )

      for (tag, guess, name) in sorted(errors):
          print('correct={:<8} guess={:<8s} name={:<30}'.format(tag, guess, name))
```

correct=female	guess=male	name=Abigail
correct=female	guess=male	name=Abigael
correct=female	guess=male	name=Adriaens
correct=female	guess=male	name=Aime
correct=female	guess=male	name=Alis
correct=female	guess=male	name=Allis
correct=female	guess=male	name=Anais
correct=female	guess=male	name=Ashleigh
correct=female	guess=male	name=Aubrey
correct=female	guess=male	name=Barb
correct=female	guess=male	name=Berry
correct=female	guess=male	name=Blakeley
correct=female	guess=male	name=Carley
correct=female	guess=male	name=Clair
correct=female	guess=male	name=Con
correct=female	guess=male	name=Cordy
correct=female	guess=male	name=Dael
correct=female	guess=male	name=Dory
correct=female	guess=male	name=Easter
correct=female	guess=male	name=Elisabet
correct=female	guess=male	name=Eran
correct=female	guess=male	name=Esme
correct=female	guess=male	name=Evangelin
correct=female	guess=male	name=Evey
correct=female	guess=male	name=Farand
correct=female	guess=male	name=Grayce
correct=female	guess=male	name=Gwendolin
correct=female	guess=male	name=Gwenn
correct=female	guess=male	name=Hester
correct=female	guess=male	name=Isabeau
correct=female	guess=male	name=Jorry
correct=female	guess=male	name=Joyan
correct=female	guess=male	name=Lesley
correct=female	guess=male	name=Lilias
correct=female	guess=male	name=Lory
correct=female	guess=male	name=Lynnett
correct=female	guess=male	name=Madel
correct=female	guess=male	name=Maren
correct=female	guess=male	name=Maridel

correct=female	guess=male	name=Maris
correct=female	guess=male	name=Maryjo
correct=female	guess=male	name=Maud
correct=female	guess=male	name=Meris
correct=female	guess=male	name=Midge
correct=female	guess=male	name=Mikako
correct=female	guess=male	name=Nance
correct=female	guess=male	name=Orel
correct=female	guess=male	name=Prue
correct=female	guess=male	name=Robbin
correct=female	guess=male	name=Rochell
correct=female	guess=male	name=Ruth
correct=female	guess=male	name=Scarlet
correct=female	guess=male	name=Scarlett
correct=female	guess=male	name=Sherry
correct=female	guess=male	name=Storm
correct=female	guess=male	name=Suzan
correct=female	guess=male	name=Suzy
correct=female	guess=male	name=Taffy
correct=female	guess=male	name=Vivyan
correct=female	guess=male	name=Winifred
correct=female	guess=male	name=Yoko
correct=male	guess=female	name=Addie
correct=male	guess=female	name=Andie
correct=male	guess=female	name=Barrie
correct=male	guess=female	name=Bartel
correct=male	guess=female	name=Bertie
correct=male	guess=female	name=Clarance
correct=male	guess=female	name=Davidde
correct=male	guess=female	name=Dennie
correct=male	guess=female	name=Durante
correct=male	guess=female	name=Elwyn
correct=male	guess=female	name=Gabriele
correct=male	guess=female	name=Georgia
correct=male	guess=female	name=Jessee
correct=male	guess=female	name=Jimmy
correct=male	guess=female	name=Kermie
correct=male	guess=female	name=Lonnie
correct=male	guess=female	name=Maddie
correct=male	guess=female	name=Morty
correct=male	guess=female	name=Nikki
correct=male	guess=female	name=Noah
correct=male	guess=female	name=Pascale
correct=male	guess=female	name=Patsy
correct=male	guess=female	name=Percy
correct=male	guess=female	name=Ronny
correct=male	guess=female	name=Ruby
correct=male	guess=female	name=Sergei

correct=male	guess=female	name=Slade
correct=male	guess=female	name=Smith
correct=male	guess=female	name=Sterne
correct=male	guess=female	name=Steve
correct=male	guess=female	name=Tally
correct=male	guess=female	name=Theodore
correct=male	guess=female	name=Toddie
correct=male	guess=female	name=Tommy
correct=male	guess=female	name=Waverly
correct=male	guess=female	name=Winnie
correct=male	guess=female	name=Zacharia

0.1.13 MODULE 12: Gender classifier improvement #4.

To make several improvements to the gender classifier, I consulted an article published by Slepian & Galinsky (2016) in the Journal of Personality and Social Psychology. They published an article, *The Voiced Pronunciation of Initial Phonemes Predicts the Gender of Names*. The authors explored the systematic effect of phonetic qualities of given names by investigating the link between sound symbolism and gender. They proposed that vocal cord vibration during the pronunciation of an initial phoneme plays a critical role in explaining which names are assigned to males versus females. This produces a voiced gendered name effect, whereby voiced phonemes (vibration of the vocal cords) are more associated with male names, and unvoiced phonemes (no vibration of the vocal cords) are more associated with female names.

The article gives examples of how one can distinguish between a voiced or unvoiced sound. To experience this difference, the authors suggest the reader pronounce the words “this” and “thin” aloud while placing a finger on the laryngeal prominence (i.e., the “Adam’s apple”). The “th” sound in this is voiced, whereas the “th” sound in thin is unvoiced. Or, pronounce the words “bear” and “pear.” One should notice in both examples (during the first phoneme) a vibration present in the former words (“this,” “bear”), but not the latter words (“thin,” “pear”), with the initial phonemes in the latter words sounding more breathy.

In Chapter 4.2 of the Bird et al text, they discuss that the NLTK library includes the CMU Pronouncing Dictionary for US English. This appears to be a bit of a complex process, so I opted to follow the Slepian & Galinsky method to distinguish voiced v. unvoiced sounds. I went through the alphabet, pronouncing each letter to classify it as voiced or unvoiced.

I modified the feature extractions to now learn from the likely female or male last letter of the name, and the voiced or unvoiced likely male or female last letter of the name.

The output suggests that the most informative features are that names ending in “na” or “ia” are nearly 9 times more likely to be female. This classifier also increased performance on the dev_set to 82.2%, up from 80.4% without the Slepian & Galinsky (2016) features.

```
[18]: def gender_features(word):
      word = word.lower()

      return {
          'voiced_ends_with_bdgvezjlmnryw': word[-1] in 'bdgvzjlmnryw',
          'unvoiced_end_with_ptkfish': word[-1] in 'ptkfish',
```

```

        'last_two_letters': word[-2:],
        'last_three_letters': word[-3:],
        'first_letter': word[0],
        'length': len(word),
    }
print(gender_features("Mary"))

```

```

{'voiced_ends_with_bdgvezjlmnryw': True, 'unvoiced_end_with_ptkfish': False,
'last_two_letters': 'ry', 'last_three_letters': 'ary', 'first_letter': 'm',
'length': 4}

```

```

[19]: train_set = [(gender_features(n), gender) for (n, gender) in train_names]
dev_set = [(gender_features(n), gender) for (n, gender) in dev_names]
classifier = nltk.NaiveBayesClassifier.train(train_set)
print(nltk.classify.accuracy(classifier, dev_set))

```

0.822

```

[20]: classifier.show_most_informative_features(5)

```

Most Informative Features

last_two_letters = 'na'	female : male	=	92.8 : 1.0
last_two_letters = 'ia'	female : male	=	88.7 : 1.0
last_two_letters = 'la'	female : male	=	71.0 : 1.0
last_two_letters = 'sa'	female : male	=	34.8 : 1.0
last_two_letters = 'ra'	female : male	=	34.2 : 1.0

```

[21]: errors = []
for (name, tag) in dev_names:
    guess = classifier.classify(gender_features(name))
    if guess != tag:
        errors.append( (tag, guess, name) )

for (tag, guess, name) in sorted(errors):
    print('correct={:<8} guess={:<8s} name={:<30}'.format(tag, guess, name))

```

correct=female	guess=male	name=Abigail
correct=female	guess=male	name=Abigael
correct=female	guess=male	name=Adriaens
correct=female	guess=male	name=Aime
correct=female	guess=male	name=Alis
correct=female	guess=male	name=Allis
correct=female	guess=male	name=Ashleigh
correct=female	guess=male	name=Aubrey
correct=female	guess=male	name=Barb
correct=female	guess=male	name=Berry
correct=female	guess=male	name=Blakeley
correct=female	guess=male	name=Carey
correct=female	guess=male	name=Carley

correct=female	guess=male	name=Clair
correct=female	guess=male	name=Con
correct=female	guess=male	name=Cordy
correct=female	guess=male	name=Dael
correct=female	guess=male	name=Dory
correct=female	guess=male	name=Easter
correct=female	guess=male	name=Eran
correct=female	guess=male	name=Esme
correct=female	guess=male	name=Evangelin
correct=female	guess=male	name=Evey
correct=female	guess=male	name=Farand
correct=female	guess=male	name=Grayce
correct=female	guess=male	name=Gwendolin
correct=female	guess=male	name=Gwenn
correct=female	guess=male	name=Hester
correct=female	guess=male	name=Isabeau
correct=female	guess=male	name=Jorry
correct=female	guess=male	name=Joyan
correct=female	guess=male	name=Lesley
correct=female	guess=male	name=Lilias
correct=female	guess=male	name=Lory
correct=female	guess=male	name=Lynnett
correct=female	guess=male	name=Madel
correct=female	guess=male	name=Maren
correct=female	guess=male	name=Maridel
correct=female	guess=male	name=Maud
correct=female	guess=male	name=Midge
correct=female	guess=male	name=Orel
correct=female	guess=male	name=Prue
correct=female	guess=male	name=Robbin
correct=female	guess=male	name=Rochell
correct=female	guess=male	name=Ruth
correct=female	guess=male	name=Scarlet
correct=female	guess=male	name=Scarlett
correct=female	guess=male	name=Sherry
correct=female	guess=male	name=Storm
correct=female	guess=male	name=Suzan
correct=female	guess=male	name=Suzy
correct=female	guess=male	name=Taffy
correct=female	guess=male	name=Tracy
correct=female	guess=male	name=Vivyan
correct=female	guess=male	name=Winifred
correct=male	guess=female	name=Addie
correct=male	guess=female	name=Andie
correct=male	guess=female	name=Barrie
correct=male	guess=female	name=Bertie
correct=male	guess=female	name=Clarence
correct=male	guess=female	name=Davidde

correct=male	guess=female	name=Dennie
correct=male	guess=female	name=Durante
correct=male	guess=female	name=Elwyn
correct=male	guess=female	name=Gabriele
correct=male	guess=female	name=Georgia
correct=male	guess=female	name=Gerrit
correct=male	guess=female	name=Jessee
correct=male	guess=female	name=Jimmy
correct=male	guess=female	name=Kermie
correct=male	guess=female	name=Kris
correct=male	guess=female	name=Lonnie
correct=male	guess=female	name=Loren
correct=male	guess=female	name=Maddie
correct=male	guess=female	name=Morty
correct=male	guess=female	name=Nikki
correct=male	guess=female	name=Pascale
correct=male	guess=female	name=Patsy
correct=male	guess=female	name=Percy
correct=male	guess=female	name=Sergei
correct=male	guess=female	name=Slade
correct=male	guess=female	name=Sterne
correct=male	guess=female	name=Steve
correct=male	guess=female	name=Tally
correct=male	guess=female	name=Theodore
correct=male	guess=female	name=Toddie
correct=male	guess=female	name=Waverly
correct=male	guess=female	name=Winnie
correct=male	guess=female	name=Zacharia

0.1.14 MODULE 13: Test Set Performance.

The code below evaluates the test set. The results are a 67% performance rate. This is unexpected. It is a significant drop from the dev_set. I speculate that this is a case of overfitting. This could be caused by being too specific in the extraction features, such as we see in the Slepian & Galinsky (2016) features, or possibly features involving the last three letters of the names.

```
[22]: print(nltk.classify.accuracy(classifier, test_set))
```

```
0.67
```